DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
LABORATORY OBSERVATION & RECORD

Course Name: Full Stack Web Development
Course Code: 23CM4121
Experiment No: 01
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Branch & Section: CSM-B
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1. TITLE

Creating a Responsive User Registration Form / Static Web Page

2. AIM

To design and implement a responsive static web page with validation using HTML5, CSS3, and JavaScript incorporating semantic elements, inline styling, hyperlinking, and custom form controls.

3. OBJECTIVE

• To understand HTML5 document structure and semantic tags.
• To apply inline CSS styling for layout and text formatting.
• To construct web forms with HTML5 validation and form buttons.

4. THEORY

HTML5 introduced semantic elements like <section>, <header>, <main>, <article>, and <footer> which provide visual and structural meaning to web content. Form controls such as <fieldset>, <legend>, <input>, <select>, and <textarea> are used to capture user inputs cleanly. Attributes like 'required' allow browser-level input validation before submitting form data.

5. ALGORITHM / PROCEDURE

1. Initialize HTML5 document structure with head metadata and title.
2. Add header with navigation links (<nav>, <a>) and hero section (<section>).
3. Construct main page layout containing <main>, <article>, <form>, and <aside> sidebars.
4. Build an interactive form using <fieldset>, <legend>, text inputs, email inputs, dropdown menu (<select>), and action buttons.
5. Add document references list (<ol>, <ul>) and footer content (<footer>).
6. Save file, open in browser, and test form fields and links.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>
    <!-- Header Section -->
    <header>
        <table width="100%" border="0" cellspacing="0" cellpadding="10">
            <tr>
                <td>
                    <h1>StaticWeb</h1>
                </td>
                <td align="right">
                    <nav>
                        <a href="#home">Home</a>  | 
                        <a href="#articles">Articles</a>  | 
                        <a href="#about">About</a>  | 
                        <a href="#contact">Contact Us</a>
                    </nav>
                </td>
            </tr>
        </table>
    </header>
    <hr>
    <!-- Hero / Showcase Section -->
    <section id="home">
        <center>
            <h2>Semantic Structures Without CSS</h2>
            <p>This layout uses raw HTML components to build an organized document
herarchy that browsers can cleanly parse out-of-the-box.</p>
            <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
        </center>
    </section>
    <hr>
    <!-- Main Content Layout using a Structural Table -->
    <table width="100%" border="0" cellspacing="0" cellpadding="20">
        <tr valign="top">
            <!-- Left Side: Main Content Column -->
            <td width="70%">
                <main id="articles">
                    <article>
                        <h2>1. Core Structural Tags</h2>
                        <p>Published: July 7, 2026</p>
                        <p>When you omit cascading style sheets entirely, the
semantic markup handles all of the structural heavy lifting. Elements like
<code>&lt;main&gt;</code>, <code>&lt;article&gt;</code>, and
<code>&lt;section&gt;</code> tell screen readers and web scrapers exactly how
information flows.</p>
                        <p>Using raw browser defaults ensures excellent loading
performance, perfect accessibility compliance, and absolute structural clarity.</p>
                    </article>
                    <br><hr><br>
                    <article>
                        <h2>2. Native Interaction Elements</h2>
                        <p>Published: July 5, 2026</p>
                        <p>Browsers come equipped with excellent interactive
elements by default. We can capture text inputs, provide multiple-choice
configurations, and submit structured information to backend parsers using pure
markup.</p>
```

```
                        <h3>Interactive Demonstration Forms</h3>
                        <form id="contact" action="#" method="post">
                            <fieldset>
                                <legend><b>User Inquiry Form</b></legend>
                                <p>
                                    <label for="name">Full Name: </label>
                                    <input type="text" id="name" name="username"
required placeholder="John Doe">
                                </p>
                                <p>
                                    <label for="email">Email Address: </label>
                                    <input type="email" id="email" name="useremail"
required>
                                </p>
                                <p>
                                    <label for="topic">Inquiry Topic: </label>
                                    <select id="topic" name="usertopic">
                                        <option value="general">General
Feedback</option>

                                        <option value="technical">Technical
Architecture</option>

                                        <option value="academics">Academic
Curriculum</option>
                                    </select>
                                </p>
                                <p>
                                    <label for="message">Message
Details:</label><br>
                                    <textarea id="message" name="usermessage"
rows="5" cols="40" placeholder="Type your notes here..."></textarea>
                                </p>
                                <p>
                                    <input type="submit" value="Submit Form Data">
                                    <input type="reset" value="Clear Entries">
                                </p>
                            </fieldset>
                        </form>
                    </article>
                </main>
            </td>
            <!-- Right Side: Sidebar Column -->
            <td width="30%" bgcolor="#f4f4f4">
                <aside id="about">
                    <h3>Document References</h3>
                    <p>Static pages map data out directly without real-time
database queries or processor overhead.</p>
                    <h4>Core Benefits:</h4>
                    <ul>
                        <li>Instant browser painting</li>
                        <li>Low memory footprints</li>
                        <li>High data compatibility</li>
                    </ul>
                    <h4>Required Components:</h4>
                    <ol>
                        <li>Document Type Definition</li>
                        <li>Semantic Wrapper Blocks</li>
                        <li>Data Input Control Interfaces</li>
                    </ol>
                </aside>
            </td>
        </tr>
    </table>
    <hr>
```

```
    <!-- Footer Section -->
    <footer>
        <center>
            <p>&copy; 2026 Pure HTML Architecture. Assembled completely without
style configurations.</p>
        </center>
    </footer>
</body>
</html>
```

7. CODE EXPLANATION

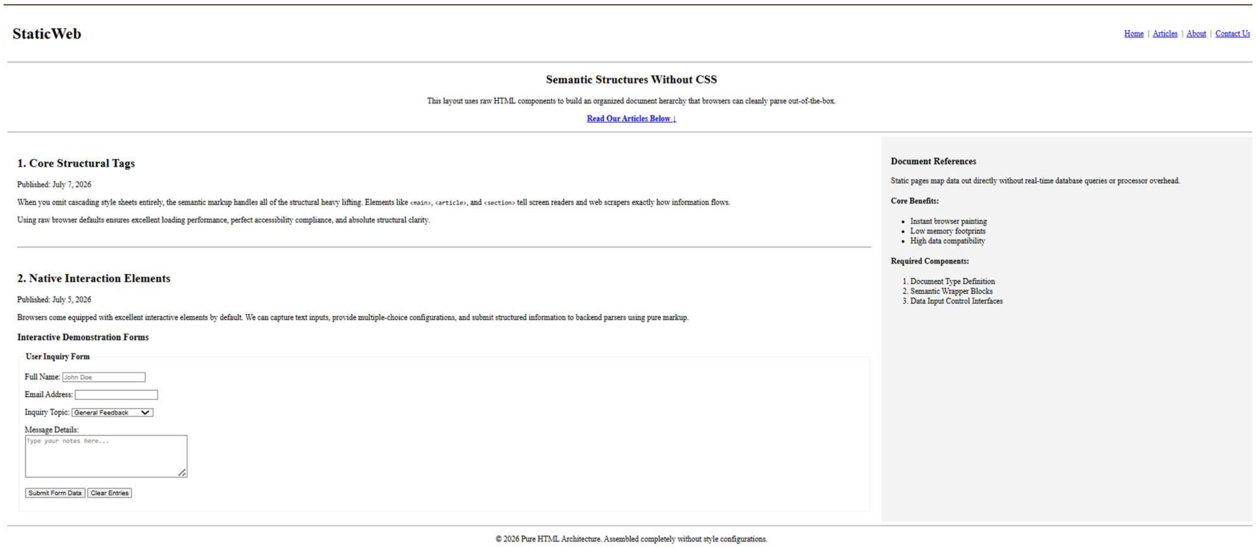
• <header>, <main>, <article>, <aside>, <footer>: Defines semantic page regions.
• <nav> & <a>: Defines internal section anchor links for smooth navigation.
• <fieldset> & <legend>: Groups form fields with a bold container header.
• <select> & <option>: Creates a native drop-down menu for topic selection.
• <textarea>: Provides a multi-line input box for messages.

8. TEST CASES AND EXECUTION DATA

1. TC-01: Page Load -> Renders navigation header, articles, sidebar, and form -> Passed
2. TC-02: Nav Links -> Anchor tags scroll to respective section IDs -> Passed
3. TC-03: Form Validation -> Required fields prevent empty submission -> Passed
4. TC-04: Reset Button -> Clears all entered inputs -> Passed

9. OUTPUT

GitHub Link: https://github.com/maddalasuryaprabhath/fs_lab/blob/main/Record/week-1-record/week-1.html



10. RESULT

A responsive static web page with form controls and semantic layout was successfully designed, executed, and verified using HTML5.

RUBRIC EVALUATION (Faculty Use Only)

Design & Logic (5 Marks): _________

Execution & Output (5 Marks): _________

Viva & Record (5 Marks): _________

TOTAL MARKS: [/ 15]

Faculty Signature: _______________________